

Labor in the Second Industrial Revolution: Substitutability and Scalability since the 1850s

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May 2026, 11th Workshop on Structural Transformation and Macroeconomic Dynamics

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Our work: We study the effects of innovations on labor via:

- (1) **substitutability: rise and fall of occupations**
 - ▷ In substitution with capital, or other occupations.
- (2) **scalability: rise or fall of employment levels (or output)**

... in the context of the largest technological disruption ever observed in human history: the US during the **Second Industrial Revolution (from mid-1800s)**.

What We Do

- Build **new database: arrival & diffusion of innovations** across time & space:
 - ▷ Source: Digitized *local* newspapers (more than 21 million pages) in the U.S. (1820-1960) (Library of Congress)
 - ▷ *Equipment ads* with pricing and *job ads* (high freq., more granular than HISCO).
 - ▷ Complemented with occupations from Population Census Dataand document the **response of occupations (and labor) to new technology**:
 - ▷ Staggered DID (Time $\times$ Space Diffusion in LOC and POP Census) (ATT Effects)
 - ▷ Shift-Share Instrument (Local ATE)
- Complement with Manufacturing Census: **wages, productivity and labor share.**
- Build a model that rationalizes our empirical results and asses short- and long-run effects (including the impact of adaptability) (TBC)

What We Find

- (1) **New technologies (1850–1950) causally imply:**
 - ▷ **Heterogenous effects (occupations): Fall** (e.g coachmen) **and rise** (e.g drivers).
 - ▷ **Increase in employment** (*Population Census* and *Job ads*)
 - ▷ **Increase in output, productivity, employment, wages and investment** (*Manufacturing Census*)
 - ▷ **labor share is unaffected** (*Manufacturing Census*)
- (2) Long-run: Job ads (demand) respond faster than employment (Census): (Lack of) adaptability?
 - ▷ Counties with higher adaptability have better outcomes in the SR and LR.
- (3) Theoretical framework: nested CES framework will allow us to estimate patterns of complementarity and substitutability across old and new labor and capital types, estimate AEOS.

Overall: new technologies reshape production across occupations and, ultimately, scale labor rather than displace it.

A New Historical Database of Technology and Occupations

A New Database

We use three data sources:

1. **Library of Congress: Chronicling America**

- Public domain repository of over 21 million newspaper pages published between 1756 and 1963.
- National coverage.
- ▷ We extract approximately 400 thousand pages containing equipment ads and 550 thousand pages containing help wanted ads.

2. **Population Census:** Detailed information on occupations and industry

3. **Manufacturing Census:** 1860-1960: Data on wages, productivity and labor share.

Data Construction: Arrival of Innovations

1. Start from the CHAT technology list of Comin and Hobijn, modifying the set of technologies to match the historical period covered by the data.
2. Construct 150 spatial regions by clustering historical Census county coordinates using K-means.
3. Geolocate newspapers using their title and publication location. For example, *The Minneapolis Journal* (Minneapolis, MN) is assigned to Minneapolis coordinates.
4. Assign each newspaper to the nearest county-based region centroid.
5. Search newspaper text for the first equipment advertisement, or alternatively the first newspaper mention, of each technology.
6. For each technology and region, define arrival as the earliest observed mention among newspapers assigned to that region.

Why this Time Period?

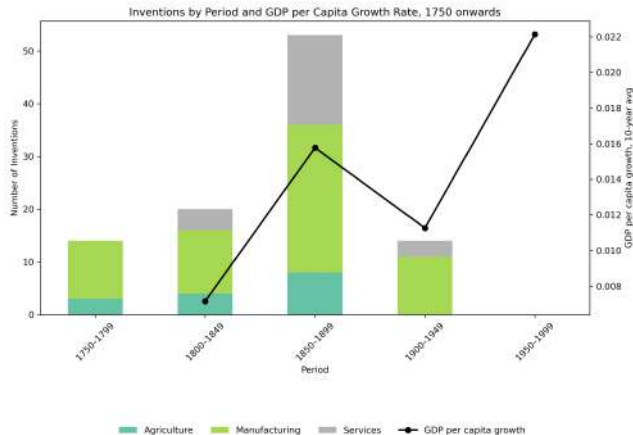


Figure 1: Number and sector of innovations? for 50 year time periods shown together with the average GDP per capita growth rate. Data sources: data on inventions is from author's calculations using the Library of Congress (LOC), data on GDP per capita is from Maddison 2023.

Tracking Technology Diffusion through Equipment Ads: Replacing Brawn

One Man Saws 40 Cords a Day

At a Cost of 1¢ a Cord! Send today for Big Special Offer and Low Direct Price on the OTTAWA, the One-Man Saw, the first made and sold direct from factory to user. Greatest labor saver and money-maker ever invented. Saws any size log at the rate of a foot a minute. Does the work of ten men. As easily moved from log to log or cut to cut as any wheelbarrow. 4-Cycle Frost Proof Engine has balanced crank shaft—pulls over 4 H-P. Magneto equipped; no batteries needed. **Special Clutch** lever controlled enables you to start and stop saw with engine running. Automatic Speed Governor. Easy to move, costs less to operate. When not sawing, engine runs pumps, feed mills and other machinery. Pulley furnished.

OTTAWA LOG SAW
Cuts Down Trees—Saws Logs By Power
Patent Applied For

Strictly a One-Man Outfit

FREE Write at once for Free Book and Low Price.

Cash or Easy Payments Shipped direct from factory. No waiting—no delay. Let the OTTAWA saw your logs and pay for itself as you use it. **10-YEAR GUARANTEE.** See the OTTAWA at work on your farm once and you will never give it up. Thousands in use, every owner a booster. Out-saws any other on the market. Send today for FREE BOOK and Special Offer.

OTTAWA MFG. CO., 2435 Wood St., Ottawa, Kans.

(a) Saw Replaces 10 Men

Every Ford Tractor carries a Power Shovel with this Dearborn Heavy Duty Loader—a Ford Tractor from that class the work of 40 men and shovels!

An exceptional worker might load 10 cubic feet of dirt, sand, gravel or manure in 20 minutes. This power loader will load 10 cubic feet of dirt, sand, gravel or manure in one quick sweep. It does the work of more than 40 men and shovels.

One Dearborn Loader equals 40 men with shovels

Wishes Power in Every New Class
This class-leading, money-making power tool will increase and reduce dirt, fill and rock loaded by dirt, sand and gravel, shovel and shovel, grain, manure, changing attachments makes it a handy tool, a great worker in a thousand places, will do what no other machine can do.

See for Yourself—Make This Test
1. It is the most powerful tractor loader ever made.
2. It is the most powerful tractor loader ever made.
3. It is the most powerful tractor loader ever made.
4. It is the most powerful tractor loader ever made.
5. It is the most powerful tractor loader ever made.
6. It is the most powerful tractor loader ever made.
7. It is the most powerful tractor loader ever made.
8. It is the most powerful tractor loader ever made.
9. It is the most powerful tractor loader ever made.
10. It is the most powerful tractor loader ever made.

Ford Farming
More Less Work...More Income For Less

See Your Nearest Ford Dealer For Complete Details and Demonstration. He Will Be Pleased To Show You.

(b) Loader Replaces 40 Men

Figure 2: Source: (left) The St. Mary banner. (Franklin, Parish of St. Mary, La.), November 20, 1920.; (right) Montana farmer-stockman. [volume] (Great Falls, Mont.), December 15, 1948.

Tracking Technology Diffusion through Equipment Ads: Replacing Brains

See the Wonderful Electric Loom Weaving

The Famous Moneybak Fine Black Silk

...Trade Mark...

Introducing
Long Wearing Moneybak Silk.
Making a Taffeta That Will Not Cut

Two representatives from the York Manufacturing Co. will be here to explain the working of the loom, and demonstrate the silk.

Silk weaving is a most interesting process to anyone desirous to see. Indeed, every perfection of weaving is a revelation. Long shelled fabric and machine made finished with human intelligence. One remarkable thing, all perfectly pure, strong threads in the back of every fabric, also a fabric closed to the filling, produce a fabric, long-wearing and lasting better silk.

Exhibit... Five Handsome Gowns

Made by a leading New York couturier, from:

Moneybak Fine Black Silk.

On Display in Nicollet Arcade Window.

Moneybak

You will see silk woven in our store just as they are woven in the mills. Every part used is manufactured in the loom or moneybak. ... that's the trademark position. "Moneybak" means in every part as a guaranteed indestructible fabric, which is detached from any silk web up to the high standard of perfection required.

Money Back

This wonderful loom will be on exhibition in our Silk and Dress Goods Depts. all week.

(a) Loom

Brinly Leverless Cultivator



The machine with almost human intelligence. No levers—no trouble—the one perfect riding cultivator—managed as easily as a walking cultivator—no extra weight for the horse if you ride.

If you will do justice to yourself you will own a **BRINLY LEVERLESS.**

FRAUENTHAL & SCHWARZ

.. Your Credit Is Good. ..

(b) Cultivator

Figure 3: Source: (left) The Minneapolis journal. [volume] (Minneapolis, Minn.), 01 March 1904.; (right) The log cabin Democrat. [volume] (Conway, Ark.), 02 March 1916.

Extracting Occupations from Help-Wanted Ads

Goal: convert unstructured help-wanted text into occupation counts that can be compared to OCC1950 Census categories.

Why trigger words are useful

Job ads often describe the work rather than using the exact Census occupation title.

*"Wanted: experienced man for machine shop.
Must operate lathe and milling machine.
Steady employment."*

The ad does not need to say *machinist* explicitly. Instead, words such as *lathe*, *milling machine*, and *machine shop* reveal the occupation.

Example classification

{*lathe*, *milling machine*, *machine shop*} $\Rightarrow$ Machinists

Procedure

- ▷ Start from OCC1950 occupation categories.
- ▷ For each occupation, define a list of related trigger words or phrases.
- ▷ Flag an occupation when the ad contains the occupation name or one of its associated triggers.
- ▷ Aggregate classified ads by occupation, year, and newspaper location.

Example: Mapping an Innovation to Exposed Occupations

Innovation: Electric Light Bulb

Prompt: For the innovation “Electric Light Bulb,” out of the following list of occupations (OCC1950 categories), list 30 which could have been affected, either negatively or positively, by the innovation in question?

Response:

Electricians; Chemical Engineers; Civil Engineers; Electrical Engineers; Mechanical Engineers; Metallurgical and Metallurgists Engineers; Mining Engineers; Engineers, n.e.c.; Photographers; Radio Operators; Medical and Dental Technicians; Testing Technicians; Technicians, n.e.c.; Managers and Superintendents, Building; Telegraph Operators; Electrotypers and Stereotypers; Linemen and Servicemen, Telegraph, Telephone, and Power; Locomotive Engineers; Machinists; Mechanics and Repairmen, n.e.c.; Motion Picture Projectionists; Stationary Engineers; Electricians Apprentices; Machinists and Toolmakers Apprentices; Mechanics except Auto Apprentices; Plumbers and Pipe Fitters Apprentices; Power Station Operators; Welders and Flame Cutters; Operative and Kindred Workers, n.e.c.

Census Data: Workers and Manufacturing Outcomes

We combine two Census-based sources to measure local labor markets and manufacturing activity.

- **Population Census / IPUMS**

- ▷ Used to measure county-level employment by occupation.
- ▷ Occupations are harmonized to the 1950 Census classification, *occ1950*.
- ▷ Occupation and industry data are available from 1850 onward.

- **Census of Manufactures / Haines (2010)**

- ▷ County-level manufacturing aggregates from *Historical, Demographic, Economic and Social Data: The US, 1790–2002*.
- ▷ Used to construct labor productivity, output, employment, average wages, capital labor ratio and labor share.

For identification we use variation *across time and space* in the arrival of innovations, and explore occupational dynamics (fall and rise).

▷ Variation across time:

- Innovation diffusion over time (equipment ads).
- Innovation-related occupations (job ads) over time.

▷ Variation across time and space:

- Innovation diffusion (equipment ads) across time and space
- Combine diffusion of innovations (equipment ads) across time and space and occupations (job ads)

Innovation Diffusion across Time (Equipment Ads)

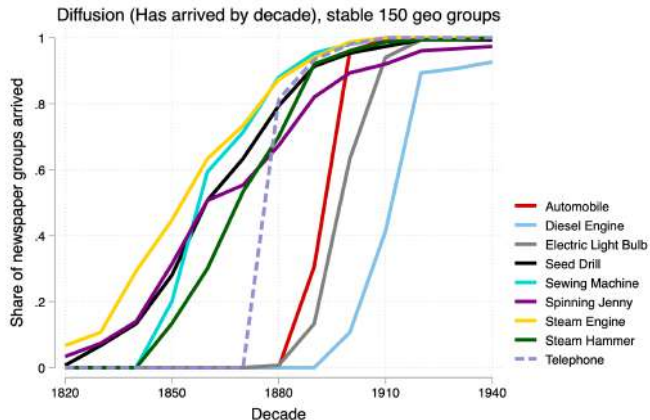


Figure 4: Fraction of newspaper clusters where a technology has appeared for a selected group of nine technologies. Source: Author's calculations using Library of Congress, *Chronicling America*.

Diffusion of Automobile Across Time and Space (Equipment Ads)

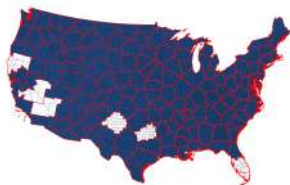
1885



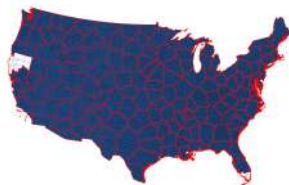
1893



1901



1909



Automobile-Related Occupations Across Time (Job Ads)

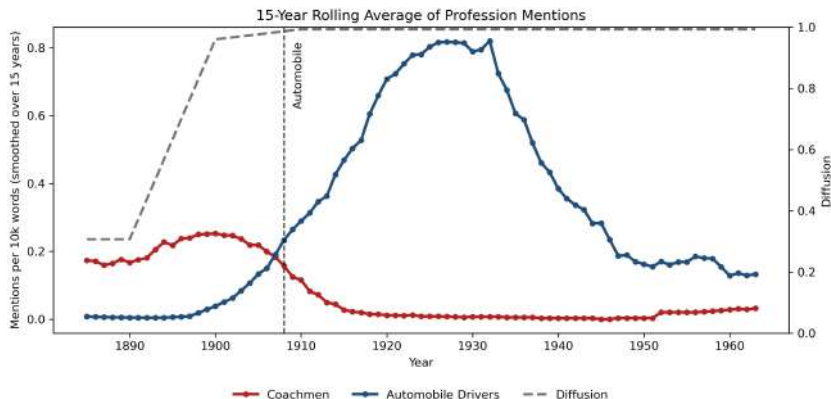


Figure 5: Automobile diffusion and occupational demand in help-wanted ads. We plot the 15-year rolling average of occupation mentions per 10,000 words in help-wanted advertisements for automobile (blue) and horse economy (red) drivers. The dashed line shows the diffusion of the automobile over time.

Diffusion of the Sewing Machine Across Time and Space (Equipment Ads)



1846



1854



1862



1870

Sewing Machine-Related Occupations (Job Ads)

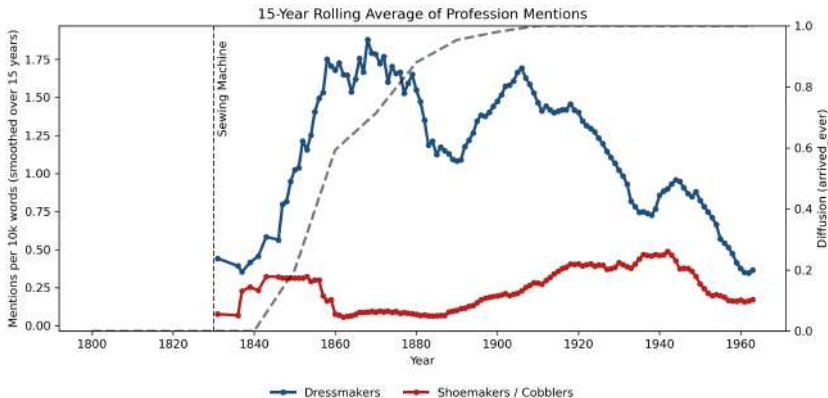


Figure 6: Sewing machine diffusion and occupational demand in help-wanted ads. We plot the 15-year rolling average of occupation mentions per 10,000 words in help-wanted advertisements for dressmakers and shoemakers/cobblers. The grey dashed line shows the diffusion of the sewing machine over time.

Diffusion of the Electric Light Bulb Across Time and Space (Equipment Ads)



1879



1887



1895



1903

Electric Light Bulb-Related Occupations (Job Ads)

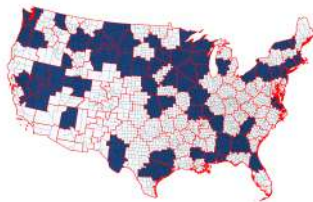


Figure 7: Electric light bulb diffusion and occupational demand in help-wanted ads. We plot the 15-year rolling average of occupation mentions per 10,000 words in help-wanted advertisements for electrical engineers and chemical engineers. The grey dashed line shows the diffusion of the electric light bulb over time.

Diffusion of the Diesel Engine Across Time and Space (Equipment Ads)



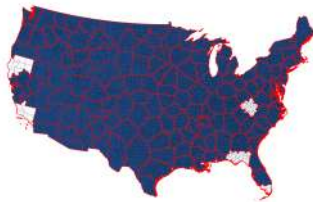
1897



1905



1913



1921

Diesel Engine-Related Occupations (Job Ads)

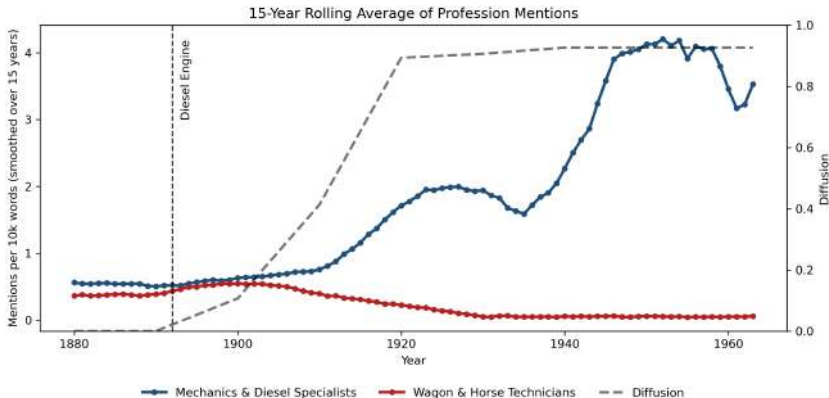
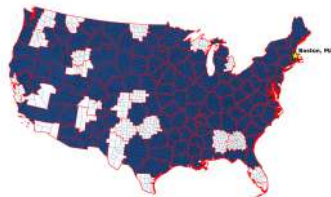


Figure 8: Diesel engine diffusion and occupational demand in help-wanted ads. We plot the 15-year rolling average of occupation mentions per 10,000 words in help-wanted advertisements for mechanics/diesel specialists and wagon/horse technicians. The grey dashed line shows the diffusion of the diesel engine over time.

Diffusion of the Telephone Across Time and Space (Equipment Ads)



1876



1884



1892



1900

Telephone-Related Occupations (Job Ads)

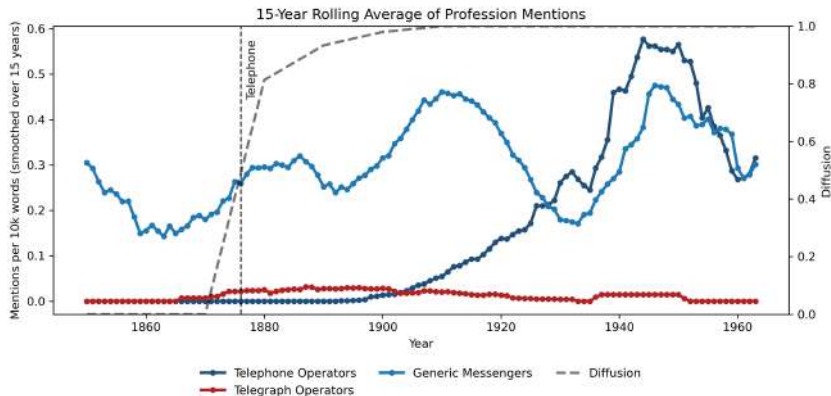
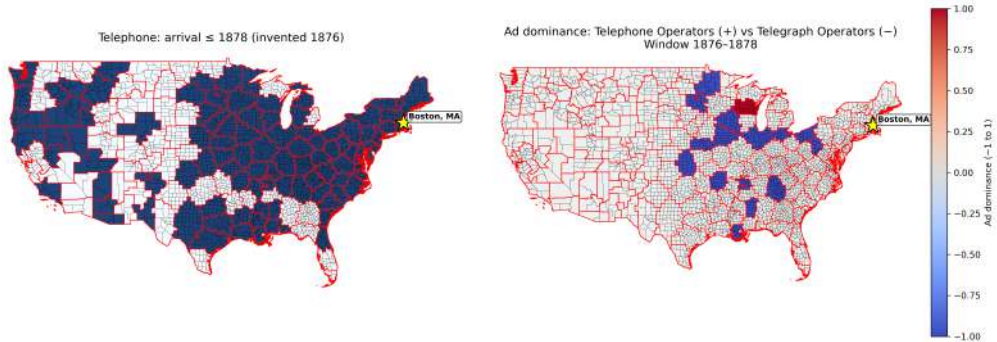


Figure 9: Telephone diffusion and occupational demand in help-wanted ads. We plot the 15-year rolling average of occupation mentions per 10,000 words in help-wanted advertisements for telephone operators, telegraph operators, and generic messenger occupations. The grey dashed line shows the diffusion of the telephone over time.

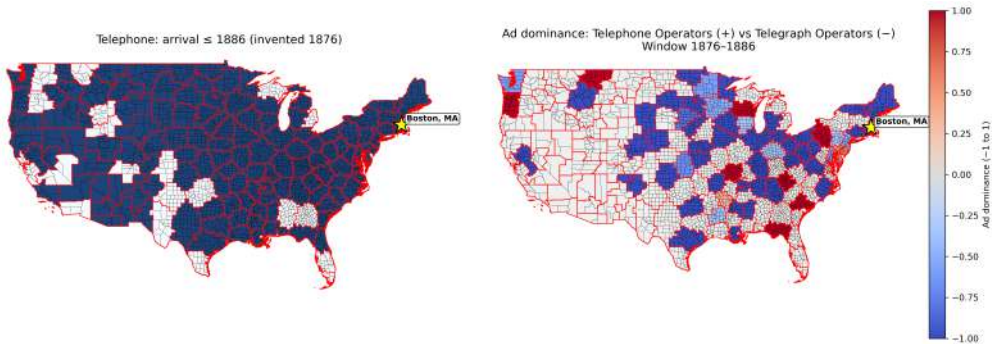
Diffusion of Telephone and Related Jobs Across Time and Space: 2 yrs

(a) Diffusion of Telephone (*Equipment Ads*) (b) Telephone Operators vs. Related Occupations (*Job Ads*)



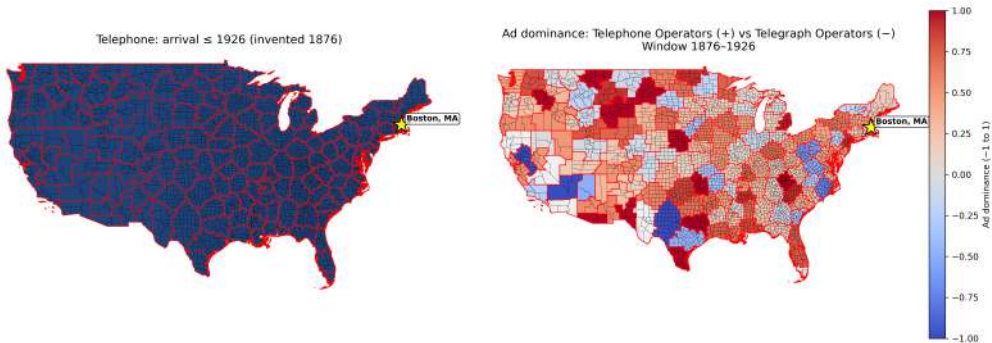
Diffusion of Telephone and Related Jobs Across Time and Space: 10 yrs

(a) Diffusion of Telephone (*Equipment Ads*) (b) Telephone Operators vs. Related Occupations (*Job Ads*)



Diffusion of Telephone and Related Jobs Across Time and Space: 50 yrs

(a) Diffusion of Telephone (*Equipment Ads*) (b) Telephone Operators vs. Related Occupations (*Job Ads*)



Empirical Effects of Innovations on Occupations: Staggered DID

Benchmark Empirical Strategy: Staggered Innovation Diffusion

We are interested in estimating the following regression model:

$$y_{nt} = \alpha_n + \delta_t + \sum_{k=-2}^2 \beta_k D_{t > \tau_n - k} + \epsilon_{nt}. \quad (1)$$

Where y_{nt} is a newspaper-cluster level outcome variable measured at the decennial level from 1850 to 1940. We use the Borusyak et al. (2024) estimator to account for staggered treatment.

Identification Assumptions: We estimate an ATT, where treatment is interpreted as the local population being aware of the existence of the innovation, as measured by its presence in local newspapers. The relevant assumptions are parallel trends and no anticipation; non-random treatment timing or heterogeneous treatment effects do not pose a threat to identification.

Results

Occupation-Level Effects of Technology Diffusion (Population Census)

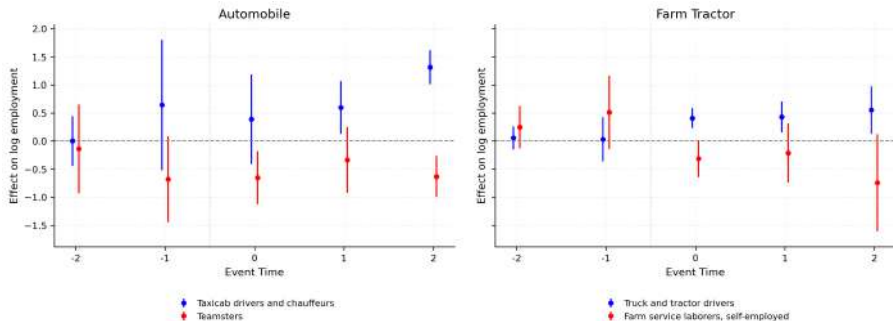


Figure 10: Event-study DiD estimates for two transport and agricultural technologies. The automobile panel compares taxicab drivers and chauffeurs with teamsters; the farm tractor panel compares truck and tractor drivers with self-employed farm service laborers. Blue marks positively affected occupations; red marks negatively affected occupations. Outcomes are log occupation employment; bars show 95 percent confidence intervals.

Occupation-Level Effects of Technology Diffusion (Population Census)

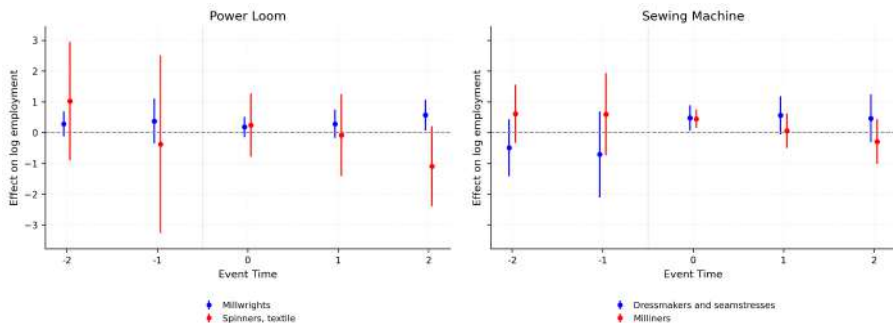


Figure 11: Event-study DiD estimates for heavy-industry and power-generation technologies. The Bessemer process panel compares engineers with blacksmiths; the steam turbine panel compares stationary engineers with forgemen and hammermen. Blue marks complementary occupations; red marks occupations with negative estimated effects. Outcomes are log occupation employment; bars show 95 percent confidence intervals.

Occupation-Level Effects of Technology Diffusion (Population Census)

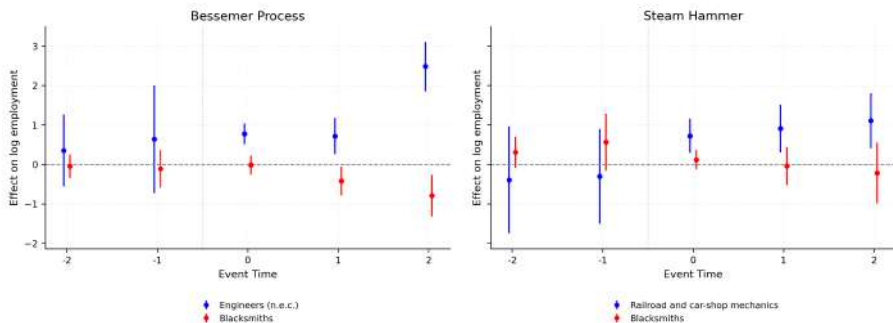


Figure 12: Event-study DiD estimates for printing and steel-production technologies. The linotype machine panel compares compositors and typesetters with photoengravers and lithographers; the Bessemer converter panel compares furnacemen, smeltermen, and pourers with rollers and roll hands. Blue denotes positive estimated effects and red denotes negative estimated effects. Outcomes are log occupation employment; bars show 95 percent confidence intervals.

ATT Heterogeneity Across Innovations (Population Census)

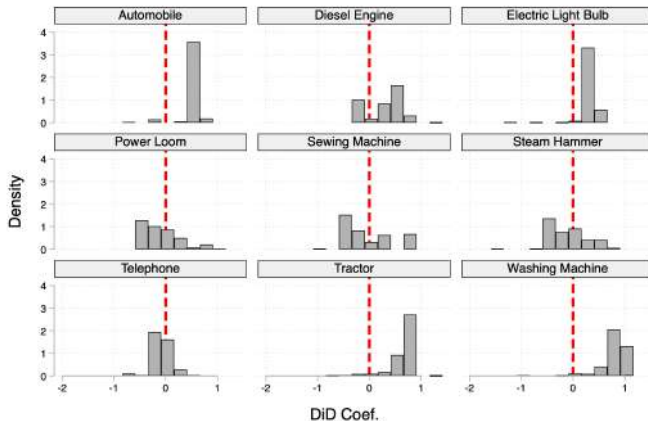


Figure 13: Distribution of estimated ATT coefficients by innovation. Each panel plots the labor-supply-weighted distribution of occupation-innovation-level treatment effects for one innovation.

ATT Heterogeneity Across Occupations (Population Census)

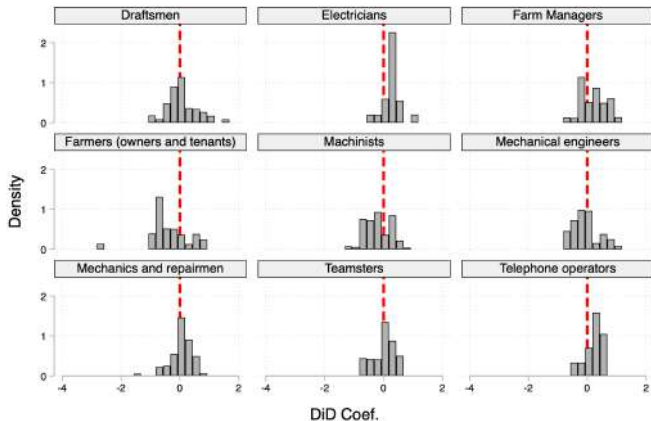


Figure 14: Distribution of estimated ATT coefficients by occupation. Each panel plots the labor-supply-weighted distribution of occupation-innovation-level treatment effects for one occupation.

Employment Effects for all Innovations×Occupations Pairs (Population Census)

Innovation-level effects		Relevant occupation-level effects	
Innovation	β	Relevant occupation	β
Automobile	0.303	Delivery/route workers	0.276
Diesel Engine	0.169	Mechanics/repairmen	0.052
Electric Light Bulb	0.133	Electricians	0.261
Farm Tractor	0.218	Truck/tractor drivers	0.302
Power Loom	0.142	Loom fixers	-0.228
Sewing Machine	-0.002	Dressmakers/seamstresses	0.320
Steam Hammer	-0.092	Forgemen/hammermen	-0.571
Telephone	-0.056	Telephone operators	0.211
All innovations / occupations	0.088	All occupations	0.088

Table 1: Unweighted average DiD effects by innovation and selected related occupations. Innovation-level effects average across occupations in each innovation crosswalk. Occupation-level effects average across all innovations in which the occupation appears.

ATTs for all Innovations×Occupations Pairs in Logs (Population Census)

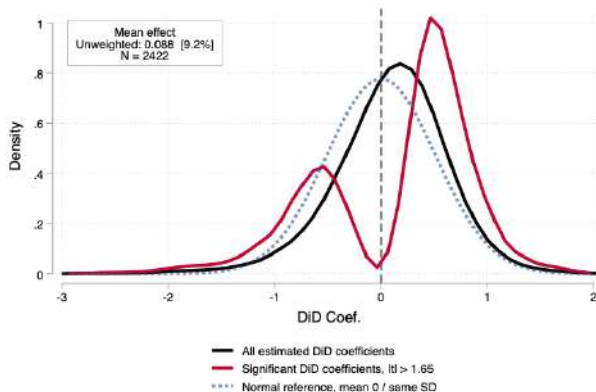


Figure 15: Distribution of estimated ATT coefficients across 2422 innovation–occupation pairs in the Population Census (30 occupations × 123 innovations, less non-identified coefficients). The black line shows the density of all estimated coefficients, while the red line restricts to statistically significant estimates ($|t| > 1.65$). The text box reports the mean effect across all estimates (black line).

ATTs for all Innovations×Occupations Pairs in Levels (Population Census)

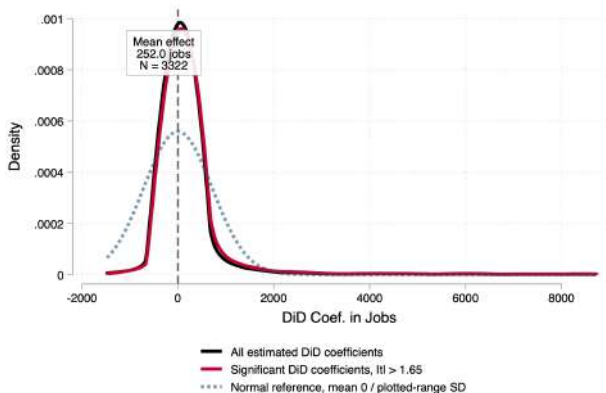


Figure 16: Distribution of estimated ATT coefficients across 3322 innovation–occupation pairs in the Population Census (30 occupations $\times$ 123 innovations, less non identified coefficients). The black line shows the density of all estimated coefficients, while the red line restricts to statistically significant estimates ($|t| > 1.65$). The text box reports the mean effect across all estimates (black line).

ATTs for all Innovations×Occupations Pairs in Logs (Job Ads)

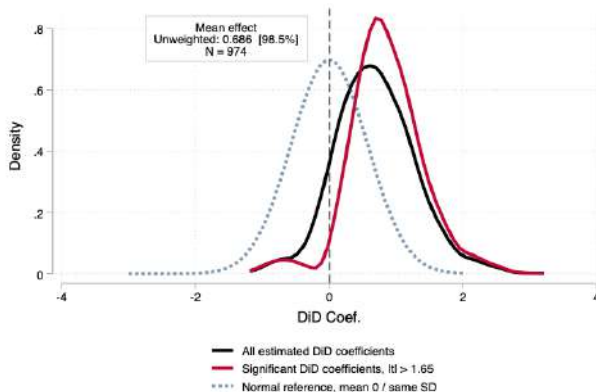


Figure 17: Distribution of estimated ATT coefficients across 974 innovation–occupation pairs in the Help Wanted Ads (30 occupations × 123 innovations, less non identified coefficients). The black line shows the density of all estimated coefficients, while the red line restricts to statistically significant estimates ($|t| > 1.65$). The text box reports the mean effect across all estimates (black line).

Dynamic Effects: Winners & Losers

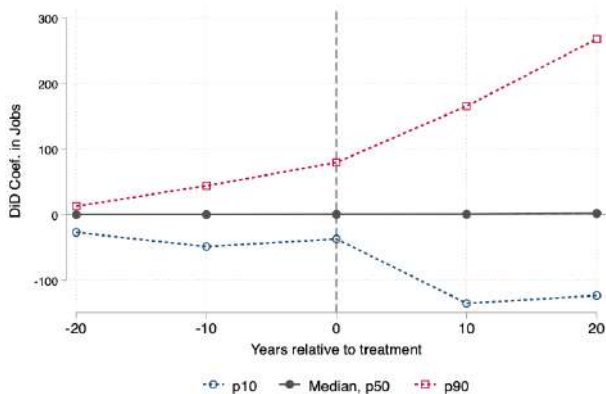


Figure 18: Distribution of dynamic DiD effects across invention–occupation cells satisfying the pre-trend test. The figure plots the 10th percentile, median, and 90th percentile of level effects over event time, where $t = 0$ is the treatment decade.

AI Classification: Heterogeneity Across Occupations

Group	Job types	Example occupations	Example effects
Winners	Agriculture; transport; infrastructure support.	Farmers; blacksmiths; brakemen; stationary firemen; lumbermen.	Steam Shovel $\times$ farmers: +26,696 Electric Furnace $\times$ blacksmiths: +2,082
Losers	Routine clerical; communication; production work.	Clerical workers; stenographers; bookkeepers; telephone operators; typesetters.	Typesetting Machine $\times$ clerical workers: -6,747 Electric Furnace $\times$ molders: -2,524
Polarized	Broad jobs exposed to both complementarity and displacement.	Machinists; laborers; farm laborers; managers; salesmen; foremen.	Machinists: -17,089 / +2,077 Salesmen: -6,130 / +7,681

Notes: Groups use pre-trend-satisfied dynamic level effects. Winners are often in the top decile, losers in the bottom decile, and polarized occupations in both tails.

Figure

Implications for Productivity, Wages and labor Share: Manufacturing Census 1850-1950

Benchmark Empirical Strategy: Manufacturing Outcomes

We estimate the effect of local innovation diffusion on county-level manufacturing outcomes using the same staggered-adoption design:

$$y_{it} = \alpha_i + \delta_t + \sum_{k=-2}^3 \beta_k \mathbf{1}\{t - \tau_i = k\} + \epsilon_{it}, \quad (2)$$

where y_{it} is a county-level manufacturing outcome measured in decade $t \in \{1860, \dots, 1940\}$, and τ_i is the decade in which an innovation first appears locally in newspapers. We then summarize the dynamic effects by taking the unweighted average of the estimated event-time coefficients across innovations. The outcomes are:

$$\log K, \quad \log(K/L), \quad \log L, \quad \log(W/L), \quad \log(W/Y), \quad \log Y, \quad \log(Y/L).$$

Effects on Employment, Output, Productivity, Wages and labor Share

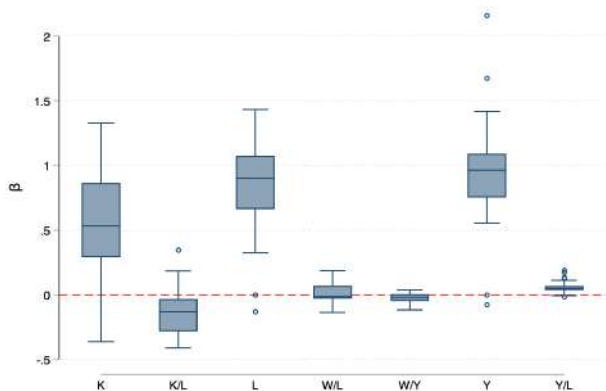


Figure 19: Distribution of the treatment effects for different outcome variables, conditional on fulfilling the pre-trend test. Each box plot shows the inter quartile range and outliers of $\hat{\beta}$ coefficients estimated from difference-in-differences regressions of model 2 using ten-year census intervals. Observations are weighted by total employment.

Dynamic Manufacturing Effects of Innovation Diffusion

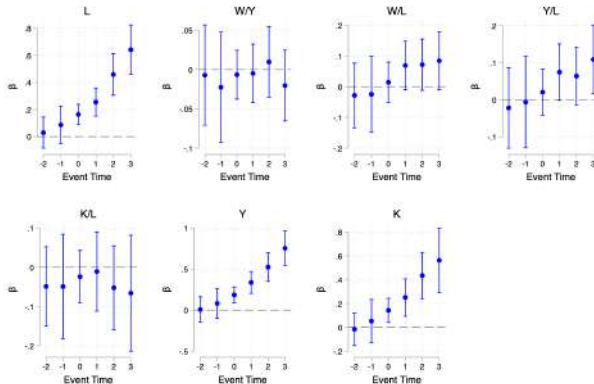


Figure 20: Average event-study effects of local innovation diffusion on manufacturing outcomes. Coefficients from model 2 are first estimated separately by innovation using the Borusyak et al. (2024) estimator, with county-decade observations weighted by labor force size, and then averaged equally across innovations. Vertical bars denote 90% confidence intervals based on average innovation-level standard errors.

Robustness: Shift-Share

Measuring Technological Exposure and Alignment

Step 1: Exposure to Innovation (Shift-Share)

- For each county c , construct exposure:

$$z_c = \sum_o s_{c,o}^{1850} \times \text{mentions}_o$$

- $s_{c,o}^{1850}$: share of workers in occupation o in 1850
- mentions_o : national intensity of innovation linked to occupation o

Step 2: Alignment between Local Demand and Supply

- Use newspaper ads to measure local labor *demand* distribution
- Use Census data to measure local labor *employment* distribution

$$\text{Alignment}_c = 1 - \frac{1}{2} \sum_o \left| \text{share}_{c,o}^{\text{ads}} - \text{share}_{c,o}^{\text{census}} \right|$$

Long-Run Effects of Innovation Exposure and Alignment

Outcome	$\log(z)$	Alignment	R^2	N
<i>Income and labor market outcomes</i>				
log income, 1999	−0.0385**	—	0.393	1,522
log income, 1999	0.0311*	1.1706***	0.455	1,522
Unemployment rate, 2000	−0.2995***	—	0.425	1,522
Unemployment rate, 2000	−0.5850***	−4.7985***	0.448	1,522
<i>Human capital and task composition</i>				
High school plus	−0.0107***	—	0.462	1,522
High school plus	0.0021	0.2139***	0.502	1,522
Abstract tasks	0.0216***	—	0.286	365
Abstract tasks	0.0204***	−0.0248	0.286	365
Manual tasks	−0.0171***	—	0.320	365
Manual tasks	−0.0169***	0.0051	0.320	365
Routine tasks	−0.0040	—	0.298	365
Routine tasks	−0.0035	0.0196	0.299	365

Notes: All regressions include state fixed effects and 1850 controls: log population, urbanization, colleges, enrollment rate, and slave population. Robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Short-Run Effects of Manufacturing Innovation, 1850–1940

Table 2: Manufacturing outcomes and manufacturing-technology exposure. All columns use the same fixed sample of 1,421 counties defined on 1850 geography.

Outcome	1850–1880		1880–1910		1910–1940	
	log(z)	s.e.	log(z)	s.e.	log(z)	s.e.
log(Y)	0.3697***	(0.0684)	0.8468***	(0.0792)	1.9694***	(0.2001)
log(L)	0.3361***	(0.0653)	0.7986***	(0.0776)	1.6505***	(0.1606)
log(Y/L)	0.0336**	(0.0146)	0.0482***	(0.0181)	0.3189***	(0.1169)
log(W/L)	0.0785***	(0.0176)	0.0587***	(0.0080)	0.2931**	(0.1158)
N	1421		1421		1421	

Notes: Unit of observation is an 1850 county. All outcome-year manufacturing variables are spatially projected to 1850 county geography using source-county area weights. All specifications include 1850 controls and state fixed effects. Robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model

Model: How Innovations Affect Labor

We model innovations as new capital, k_1 , and ask how they affect two labor groups:

- h_1 : occupations complementary to the new innovation;
- h_0 : occupations associated with older technologies.

The response of labor to innovation operates through three channels:

1. **Direct substitution/complementarity** between k_1 and h_1 ;
2. **Reallocation across labor types** between the new-technology composite and h_0 ;
3. **Scalability** through the output response y .

The goal is to use the observed responses of h_1 and h_0 to k_1 to discipline the elasticities governing these channels.

Nested CES Production Structure

The production function has three nests:

$$y = \left((1 - \theta)k_0^{\frac{\psi-1}{\psi}} + \theta q^{\frac{\psi-1}{\psi}} \right)^{\frac{\psi}{\psi-1}}, \quad (3)$$

$$q = \left((1 - \mu)g^{\frac{\sigma-1}{\sigma}} + \mu h_0^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (4)$$

$$g = \left((1 - \lambda)k_1^{\frac{\rho-1}{\rho}} + \lambda h_1^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}. \quad (5)$$

- ρ : substitutability between new capital k_1 and new-technology labor h_1 .
- σ : substitutability between the new-technology composite g and other labor h_0 .
- ψ : substitutability between old capital k_0 and the composite q .

Labor Response Decomposition

For exposition, suppress the outer nest and set $y = q$. Then the labor demand response of h_1 can be written as:

$$\frac{dh_1}{dt} = \underbrace{\frac{\partial h_1}{\partial k_1} \frac{dk_1}{dt}}_{\text{Substitution with } k_1} + \underbrace{\frac{\partial h_1}{\partial h_0} \frac{dh_0}{dt}}_{\text{Reallocation across labor types}} + \underbrace{\frac{\partial h_1}{\partial y} \frac{dy}{dt}}_{\text{Scalability}}. \quad (6)$$

Interpretation:

- holding h_0 and y fixed isolates the substitution margin between k_1 and h_1 ;
- allowing h_0 to move captures labor reallocation across nests;
- allowing y to move captures the output-scale effect.

The decomposition for h_0 is analogous.

Identifying Elasticities from Labor Responses

We identify the elasticities by inverting the nested CES structure.

For any candidate ρ , observed (k_{1t}, h_{1t}) imply a path for the new-technology composite:

$$g_t(\rho) = \left((1 - \lambda) k_{1t}^{\frac{\rho-1}{\rho}} + \lambda h_{1t}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}. \quad (7)$$

For any candidate σ , observed (y_t, h_{0t}) imply another path for the same composite:

$$g_t(\sigma) = \left[\frac{1}{1 - \mu} \left(y_t^{\frac{\sigma-1}{\sigma}} - \mu h_{0t}^{\frac{\sigma-1}{\sigma}} \right) \right]^{\frac{\sigma}{\sigma-1}}. \quad (8)$$

The identifying restriction is:

$$g_t(\rho) = g_t(\sigma) \quad \forall t. \quad (9)$$

Identification Logic and Aggregate Implications

With one period,

$$F_t(\rho, \sigma) = g_t(\rho) - g_t(\sigma) = 0 \quad (10)$$

provides one equation in two unknowns, so many (ρ, σ) pairs can rationalize the data.

With multiple periods, the elasticities are disciplined by matching the full response path:

$$(\hat{\rho}, \hat{\sigma}) = \arg \min_{\rho, \sigma} \sum_t [g_t(\rho) - g_t(\sigma)]^2. \quad (11)$$

Once identified, these elasticities map the micro labor responses into aggregate outcomes:

- the decomposition of labor responses into substitution and scalability;
- a time-varying aggregate elasticity of substitution between capital and labor.

Conclusion

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- We construct a county level dataset of technology diffusion and labor demand for the US over 1800/1960 covering 123 technologies.
- Technological innovations between 1860–1950 substituted and scaled different occupations: **reallocation and employment growth dominate displacement.**
- Average effects on labor markets are **positive on wages, productivity, and employment. labor share unaffected.**
- Set up a flexible model in which innovation-led output growth can lead to either substitutability (or complementarity) and to scalability or not.
- Long-run effects: Regions that adapted faster are richer, more polarized, and less unemployed by 2000.
- From a historical perspective, we offer a more optimistic perspective than views about current technological transitions (Acemoglu and Restrepo, 2019, 2021).